# BABAPPAΩ: a likelihood-free diagnostic for branch–site episodic-selection identifiability

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**Abstract**

Branch–site methods are widely used to infer episodic positive selection from coding-sequence alignments, but their conclusions depend on explicit codon-model assumptions and on whether the data contain enough information to resolve lineage-restricted signal. Recombination, alignment uncertainty, heterogeneous background constraint, and limited sequence information can therefore make branch–site inference statistically unstable even when biologically meaningful perturbation is present. Here I present BABAPPAΩ, a likelihood-free diagnostic framework that targets branch–site *identifiability* rather than binary detection. The method combines forward simulation under codon-level mutation–selection dynamics with amortized neural inference trained to recognize branch–site deviations from a neutral or weakly structured background, without gene-wise likelihood fitting or direct estimation of $\omega$. From the inferred branch–site scores, BABAPPAΩ derives aggregate burden summaries and an Episodic Identifiability Index (EII). To support end-user interpretation, the framework reports both a standardized effect-size statistic ($\mathrm{EII}_z$) and a bounded operational score ($\mathrm{EII}_{01} \in [0, 1]$) calibrated against an empirical neutral reference, together with regime labels ranging from *not identifiable* to *strongly identifiable*. In internal synthetic benchmarks, predicted burden increased monotonically with simulated episodic load, remained unstructured under neutrality, and was more stable at gene level than at individual sites. Performance declined under recombination-aware segmentation, heterogeneous mutation rates, and alignment noise, but the degradation was gradual rather than catastrophic. BABAPPAΩ is intended as a complement to classical branch–site tests, not a replacement: it diagnoses whether episodic inference is well posed for a given dataset, whether it is currently inconclusive, and when the available signal is strong enough to justify deeper downstream analysis.

**Keywords:** episodic selection; branch–site evolution; likelihood-free inference; simulation-based inference; mutation–selection models; phylogenomics

## 1 Introduction

Detecting adaptive evolution in protein-coding genes is a central objective of comparative genomics. Codon-based likelihood methods, especially implementations in PAML and HyPhy, have dominated this area because they provide explicit statistical models for nonsynonymous and synonymous substitution processes [16, 13, 12]. Within this framework, branch–site models are attractive because they allow elevated nonsynonymous rates at a subset of codons along prespecified lineages, matching the biological expectation that adaptation is often sparse in both time and sequence space [3, 17, 11].

Yet classical branch–site inference addresses a narrower target than is often assumed. In practice, these methods do not simply ask whether episodic selection exists in a broad biological sense; they ask whether the data support that conclusion under a specific codon-model parameterization. This distinction matters because limited alignment length, shallow or excessive divergence, recombination, and alignment error can all degrade statistical recoverability. Earlier simulation studies showed that the performance of likelihood-ratio procedures depends strongly on sequence divergence, model misspecification, recombination, and alignment quality [1, 15, 6, 10, 2]. As a result, the absence of statistical support is not equivalent to the absence of biological signal.

This gap motivates a different inferential question: *is episodic signal recoverable from the alignment at all, and at what scale?* Recent advances in simulation-based and likelihood-free inference make this question tractable. Neural networks have shown substantial utility in evolutionary and population-genetic inference because they can learn informative mappings from sequence patterns without requiring a closed-form likelihood [14, 5, 4]. This is especially useful when the simulator can encode richer biological structure than the downstream analytic model.

BABAPPAΩ was developed from this perspective. Instead of treating episodic selection as a shift in codon-model $\omega$ classes, the framework represents it as a branch- and site-specific perturbation of latent codon-fitness landscapes within a mutation–selection process [8]. A neural model is trained on simulator-generated alignments with exact latent labels to infer branch–site deviation scores from observed sequence patterns. These scores are aggregated into burden summaries and an Episodic Identifiability Index (EII), which measures whether the alignment contains enough structured variation for episodic interpretation to be statistically meaningful.

For a methods-oriented audience, the central claim must be stated carefully. BABAPPAΩ is *not* presented as a replacement for codon-model tests, a direct estimator of positive selection, or a posterior probability engine. Instead, it is a diagnostic layer that asks whether the branch–site question is statistically well posed for the dataset under study. This positioning is important because methods papers are strongest when the inferential target, the validation scope, and the practical use case are explicitly aligned.

# 2 Materials and methods

## 2.1 Conceptual overview

BABAPPAΩ consists of three components: (i) a forward simulator of coding-sequence evolution under branch-specific mutation–selection dynamics, (ii) a neural inference model trained on simulator-generated alignments with exact latent labels, and (iii) a calibration layer that summarizes branch–site scores into aggregate burden measures and EII. The framework is likelihood-free at inference time: it does not perform gene-wise likelihood optimization, branch-wise $\omega$ estimation, or likelihood-ratio testing.

## 2.2 Forward simulation under mutation–selection dynamics

Evolution is simulated on the 61-sense-codon state space under the standard genetic code, excluding stops. At each site $i \in \{1, \ldots, L\}$, a latent codon-fitness landscape is defined as

$$F_i : \mathcal{C} \to \mathbb{R},$$

where $F_i(c)$ denotes the fitness of codon $c$ at site $i$. Baseline landscapes are sampled as

$$F_i^{(0)}(c) \sim \mathcal{N}(0, \sigma_F^2),$$

allowing a continuum of weakly deleterious, nearly neutral, and mildly favorable states without imposing discrete codon-model classes.

To introduce gene-wide codon-usage heterogeneity, a codon-specific offset is added:

$$F_i(c) = F_i^{(0)}(c) + \gamma \log\left(\frac{p_c}{\bar{p}}\right),$$

where $p_c$ is sampled from a Dirichlet distribution over the codon state space, $\bar{p}$ is the mean codon frequency, and $\gamma$ modulates the contribution of codon-usage preference.

Episodic selection is generated as a sparse, branch- and site-specific perturbation of the latent fitness landscape. For selected branch–site pairs $(i, b)$,

$$F_{i,b}(c) = F_i(c) + \Delta F_{i,b}(c), \qquad \Delta F_{i,b}(c) \sim \mathcal{N}(0, \sigma_{\text{epi}}^2).$$

Only a sparse subset of branch–site pairs receives perturbation, preserving the expectation that episodic adaptation is localized in both lineage and position. To approximate local genealogical heterogeneity, the alignment can be partitioned into contiguous blocks that act as coarse recombination-aware segments.

Single-nucleotide mutations follow an HKY-type scheme with transition–transversion bias [9]. If codons $c$ and $c'$ differ by exactly one nucleotide,

$$\mu_{c \to c'} = \begin{cases} \kappa\mu, & \text{if the change is a transition,} \\ \mu, & \text{if the change is a transversion,} \end{cases}$$

where $\mu$ is the baseline mutation rate and $\kappa$ is the transition bias parameter. For branch $b$ and site $i$, the selection coefficient of a proposed change is

$$s_{i,b}(c \to c') = F_{i,b}(c') - F_{i,b}(c).$$

Substitutions are then realized under a Halpern–Bruno-style mutation–selection process [8], with continuous-time trajectories simulated using a Gillespie algorithm [7].

Because episodic perturbations are specified explicitly, exact latent labels are available. For each site–branch pair $(i, b)$, the simulator records an episodic indicator $Y_{i,b} \in \{0, 1\}$. These indicators are aggregated into site- and branch-level burdens:

$$B_i^{\text{site}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} Y_{i,b}, \qquad B_b^{\text{branch}} = \frac{1}{L} \sum_{i=1}^{L} Y_{i,b},$$

where $\mathcal{B}$ is the set of branches in the tree.

### 2.3 Neural inference and training objective

Each training example consists of a codon alignment together with branch-conditioned features derived from parent–child codon transitions, branch lengths, and positional context. The model is trained in two stages. First, it is pretrained on a large corpus of simulated alignments spanning

neutral, weakly structured, and strongly episodic regimes. Second, it is fine-tuned against aggregate burden targets so that network outputs remain aligned with simulator-defined site- and branch-level burden.

Let $\ell_{i,b}$ denote the site–branch logit predicted by the network, and let $\sigma(x) = 1/(1 + e^{-x})$ be the logistic function. The predicted site- and branch-level burdens are

$$\hat{B}_i^{\text{site}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \sigma(\ell_{i,b}), \qquad \hat{B}_b^{\text{branch}} = \frac{1}{L} \sum_{i=1}^{L} \sigma(\ell_{i,b}).$$

The fine-tuning objective is written explicitly as a burden-level binary cross-entropy on soft targets:

$$\mathcal{L}_{\text{site}} = -\frac{1}{L} \sum_{i=1}^{L} \Big[ B_i^{\text{site}} \log \hat{B}_i^{\text{site}} + \big(1 - B_i^{\text{site}}\big) \log \big(1 - \hat{B}_i^{\text{site}}\big) \Big],$$

$$\mathcal{L}_{\text{branch}} = -\frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \Big[ B_b^{\text{branch}} \log \hat{B}_b^{\text{branch}} + \big(1 - B_b^{\text{branch}}\big) \log \big(1 - \hat{B}_b^{\text{branch}}\big) \Big],$$

with total loss

$$\mathcal{L}_{\text{total}} = \lambda_s \mathcal{L}_{\text{site}} + \lambda_b \mathcal{L}_{\text{branch}},$$

where $\lambda_s$ and $\lambda_b$ are fixed nonnegative weights. This notation makes clear that the optimization acts on burden summaries rather than on a second logistic transformation of an already bounded quantity.

After training, model parameters are frozen and used as an immutable inference artifact. At inference time, BABAPPAΩ returns branch–site deviation logits, site-level burdens, branch-level burdens, gene-level summaries obtained by averaging site scores within each coding region, and EII-based interpretation labels. These outputs are diagnostic and relative; they are not posterior probabilities of positive selection and they do not replace formal codon-model likelihood statistics.

### 2.4 Episodic Identifiability Index: raw and bounded forms

The framework aims to distinguish the existence of latent episodic perturbation in the generative process from its recoverability in observed sequence data. To summarize this recoverability, BABAPPAΩ computes an Episodic Identifiability Index (EII).

For a gene with $L$ codon sites, let

$$m_i = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \ell_{i,b}$$

be the branch-averaged site logit, and let the default observed dispersion statistic be

$$D_{\text{obs}} = \text{Var}_{i=1,\ldots,L}(m_i).$$

This choice is operational rather than exclusive; other branch-aware dispersion summaries can be substituted in sensitivity analyses, but variance of branch-averaged logits is the default statistic reported by the software.

Neutral or near-neutral Monte Carlo replicates matched to the alignment configuration

generate reference statistics $D_0^{(1)}, \ldots, D_0^{(R)}$, from which

$$\mu_0 = \frac{1}{R}\sum_{r=1}^{R} D_0^{(r)}, \qquad \sigma_0 = \sqrt{\frac{1}{R-1}\sum_{r=1}^{R}\big(D_0^{(r)} - \mu_0\big)^2}.$$

The standardized effect-size form of EII is then

$$\mathrm{EII}_z = \frac{D_{\mathrm{obs}} - \mu_0}{\sigma_0}.$$

$\mathrm{EII}_z$ is useful because it retains a scale-sensitive description of how strongly the observed dispersion exceeds the matched null.

For end-user reporting, however, a bounded score is often more practical. BABAPPAΩ therefore also reports

$$\hat{p}_+ = \frac{1 + \sum_{r=1}^{R} \mathbf{1}\Big[D_0^{(r)} \geq D_{\mathrm{obs}}\Big]}{R+1},$$

which is the empirical one-sided upper-tail probability under the matched null, and defines the bounded operational score

$$\mathrm{EII}_{01} = \max\big(0,\, 1 - 2\hat{p}_+\big).$$

By construction, $\mathrm{EII}_{01} \in [0, 1]$. Values near 0 indicate that the dataset is no more structured than the neutral background for the branch–site question at hand; values near 1 indicate that the observed dispersion is more extreme than almost all matched null replicates. The software reports both $\mathrm{EII}_z$ and $\mathrm{EII}_{01}$, because the bounded score is easy to interpret operationally whereas the raw standardized score remains useful for methods development and sensitivity analysis.

### 2.5 Operational interpretation and decision regimes

To provide a definitive runtime answer for end users, BABAPPAΩ attaches an interpretation regime to each analyzed dataset based on $\mathrm{EII}_{01}$. The operational reporting scheme is

| $EII_{01}$ range | Runtime label | Interpretation |
|---|---|---|
| $0.00 \leq EII_{01} < 0.30$ | Not identifiable | The data do not contain resolvable branch–site structure beyond matched neutral expectation. |
| $0.30 \leq EII_{01} < 0.70$ | Weak / ambiguous | There is some excess structure, but it is not stable enough for a positive identifiability call. Additional taxa, longer alignments, better alignment quality, or branch redefinition may be needed. |
| $0.70 \leq EII_{01} < 0.90$ | Identifiable | The dataset is operationally identifiable for branch–site episodic structure, with moderate recoverability. Downstream codon-model analyses are justified but should still be interpreted cautiously. |
| $0.90 \leq EII_{01} \leq 1.00$ | Strongly identifiable | The dataset shows strong branch–site structure relative to the matched null and is highly suitable for downstream episodic analysis. |

For conservative software behavior, the binary decision flag is defined as

$$\texttt{identifiable} = \begin{cases} \texttt{no}, & EII_{01} < 0.70, \\ \texttt{yes}, & EII_{01} \geq 0.70. \end{cases}$$

Thus, the interval $[0.30, 0.70)$ is treated as *inconclusive rather than positive*. This rule provides a single operational answer while preserving the more informative regime labels for users who want to know the extent of identifiability.

### 2.6 Validation and benchmarking design

Validation was performed on held-out simulated datasets generated with independent random seeds and tree realizations. No alignment used for validation or benchmarking was seen during training or fine-tuning, and all results were obtained with the frozen model in evaluation mode.

Because episodic burden is fundamentally relative, primary evaluation used rank-based criteria. For each alignment, Spearman's rank correlation was computed between true and predicted site burdens and, separately, between true and predicted branch burdens. When the true burden vector was constant, as under strict neutrality, rank correlation is mathematically undefined; such cases were treated as correctly uninformative rather than as failures.

Additional diagnostics included enrichment of true signal among the highest-scoring sites, bootstrap intervals for rank summaries, and a channel-collapse index assessing whether branch-specific outputs degenerated into nearly identical internal representations. Stress tests introduced recombination-aware segmentation, heterogeneous mutation rates, alignment noise, and malformed input files.

For multi-gene synthetic benchmarks, independently simulated genes were concatenated into larger datasets with known gene-level burdens:

$$B_g^{\text{true}} = \frac{1}{L_g} \sum_{i=1}^{L_g} Y_i, \qquad B_g^{\text{pred}} = \frac{1}{L_g} \sum_{i=1}^{L_g} \hat{s}_i,$$

where $L_g$ is gene length, $Y_i$ records latent episodic membership, and $\hat{s}_i$ is the inferred site score. Four broad burden regimes were considered—neutral, low, medium, and high—to assess whether aggregation improved recoverability.

The present manuscript intentionally limits its main claims to *internal synthetic validation and stress testing*. Head-to-head comparison against codeml branch–site tests, aBSREL, MEME, and related procedures is an important next step and should be included in a full comparative benchmarking study. Likewise, empirical case studies with well-motivated candidate genes will be necessary before the framework can be judged as a broadly validated practical method rather than a promising diagnostic prototype.

# 3 Results

## 3.1 Aggregation improves recoverability in internal synthetic benchmarks

Across held-out non-neutral simulations, BABAPPAΩ recovered latent episodic burden monotonically as the fraction of perturbed sites increased. The strongest and most stable agreement with ground truth emerged at gene level, where predicted burden ranked genes according to simulated episodic load. Site-level resolution was consistently weaker, especially in low-load regimes. This scale dependence is central to the framework: episodic structure may become statistically measurable after aggregation even when individual codons do not yet carry sufficiently stable signal for precise localization.

## 3.2 Neutral alignments are correctly uninformative

Under strictly neutral simulations, true burden is constant by construction and rank-based association is undefined because no ground-truth ordering exists. In this setting, BABAPPAΩ produced low-dispersion, unstructured outputs rather than spurious burden rankings. This behavior is desirable because a diagnostic method should remain uninformative when the data contain no episodic structure to recover.

## 3.3 Robustness under biologically plausible perturbations

When recombination-aware block structure, heterogeneous mutation rates, and alignment noise were introduced, performance declined relative to the cleanest benchmarks but deteriorated gradually rather than catastrophically. Gene-level ranking remained informative over a broad range of perturbed conditions, whereas fine-scale site localization was more sensitive. This pattern is consistent with earlier observations that recombination and alignment error can distort site-wise selection inference in codon-model frameworks [2, 6, 10]. The present results therefore support the use of aggregate diagnostics before strong site-specific claims are made.

### 3.4 EII provides a bounded decision together with a graded extent label

In internal synthetic benchmarks, $\mathrm{EII}_z$ increased with simulated episodic burden and remained near the neutral baseline in negative-control datasets. The bounded companion score $\mathrm{EII}_{01}$ translated this behavior into an operational reporting scale from 0 to 1. Datasets in the low-burden and neutral regimes concentrated in the *not identifiable* or *weak/ambiguous* intervals, whereas moderate- and high-burden simulations increasingly occupied the *identifiable* and *strongly identifiable* intervals.

This two-level reporting is practically useful. First, the binary decision flag answers the direct user question: *is the dataset currently identifiable for branch–site episodic analysis or not?* Second, the regime label describes the extent of support: *not identifiable*, *weak/ambiguous*, *identifiable*, or *strongly identifiable*. Third, the retained $\mathrm{EII}_z$ value allows advanced users to inspect effect size and calibration behavior rather than relying on a bounded score alone.

### 3.5 Software behavior is deterministic and robust

Repeated runs on identical inputs produced identical outputs, indicating deterministic inference after model freezing. Invalid inputs, including empty alignments, non-codon-length sequences, and single-taxon files, were rejected with explicit error messages rather than silent corruption. Conserved negative controls generated nearly flat score profiles, whereas synthetic alignments containing localized branch-specific perturbation produced corresponding peaks in inferred burden. These checks support the practical robustness of the implementation.

## 4 Discussion

BABAPPAΩ is built around a simple but important shift in target: from *detection* of positive selection under a codon-model null to *identifiability* of episodic branch–site structure from the available alignment. These goals are related but not identical. In realistic settings, biological episodic perturbation may be present while the statistical information needed to localize or test it remains weak. A method that explicitly diagnoses this limitation can therefore add value even when classical branch–site tools remain the primary inferential engines.

A second central result is that aggregation stabilizes signal. In the present internal benchmarks, episodic structure was more recoverable at gene level than at individual sites. This is both biologically and statistically plausible: episodic adaptation is often sparse, lineage-restricted, and transient, so local evidence may be weak even when the gene as a whole carries real episodic burden. BABAPPAΩ does not eliminate this difficulty; rather, it quantifies where along the scale hierarchy recoverability begins to emerge.

The EII formulation is also clearer in its revised form. Reporting both $\mathrm{EII}_z$ and $\mathrm{EII}_{01}$ separates two roles that were previously conflated. The standardized statistic provides a methods-oriented description of excess dispersion relative to the matched neutral reference, whereas the bounded score provides a user-facing interpretation scale and a definitive runtime decision rule. From a reviewer perspective, this is important because a methods paper should distinguish the *statistical object being measured* from the *software convention used for interpretation*. The operational regime boundaries used here are intended for software reporting and reproducible practice; they should not be mistaken for universal biological constants.

The method is intended to complement, not replace, established branch–site procedures such as PAML and HyPhy [16, 13, 11, 12]. In practice, classical likelihood-based tests can still be

used to estimate selection effects under explicit codon models, but their outputs may be interpreted more cautiously when BABAPPAΩ reports low identifiability. Conversely, convergence between strong classical evidence and high EII should increase confidence that the data contain recoverable episodic structure rather than unstable model-driven signal.

Several limitations should be stated explicitly. First, the realism of inference depends on the realism of the simulator. Although the simulator incorporates mutation bias, codon-usage heterogeneity, branch-specific perturbation, and local segmentation, it cannot capture every feature of real coding-sequence evolution. Second, the current framework yields diagnostic scores rather than formal posterior probabilities or definitive causal evidence for adaptation. Third, although the operational EII regime system is useful for software reporting, its transferability across highly atypical phylogenetic regimes should be evaluated empirically. Fourth, the current validation emphasizes synthetic data and stress tests; a strong methods paper should ultimately include direct comparison with current practice and empirical demonstrations on biologically compelling case studies.

These limitations also point directly to what an MBE reviewer is likely to require in a stronger revision: cleaner formal definitions, explicit separation of internal validation from broad real-world claims, clearer comparison to current branch–site practice, and a practically interpretable decision layer that users can apply without overclaiming what the method proves. The present revision is designed to move the manuscript in that direction.

## 5 Conclusion

BABAPPAΩ reframes branch–site analysis around the recoverability of episodic signal. By combining forward simulation under mutation–selection dynamics with likelihood-free neural inference, it diagnoses whether coding-sequence alignments contain enough structured variation for episodic interpretation and identifies the scale at which that structure becomes measurable. The revised EII formulation now supports both methods-oriented calibration ($\mathrm{EII}_z$) and user-facing operational reporting ($\mathrm{EII}_{01}$ with explicit regime labels). The framework does not substitute for classical branch–site testing, but it provides a useful diagnostic layer that helps distinguish biologically absent signal from statistically unrecoverable signal.

## Availability and implementation

BABAPPAΩ is available as an open-source Python package at https://github.com/sinhakrishnendu/babappaomega. A frozen reference model for reproducible inference is archived on Zenodo at https://doi.org/10.5281/zenodo.18195868. Benchmarking artifacts, execution logs, and related reproducibility materials are archived at https://doi.org/10.5281/zenodo.18197957.

## Data availability

All benchmark data used in this study were generated by simulation. The source code, frozen inference artifacts, and archived benchmark outputs required to reproduce the analyses are available through the repository and Zenodo records listed above.

## Supplementary data

Supplementary data, including additional benchmark summaries, stress-test logs, calibration diagnostics for $\mathrm{EII}_z$ and $\mathrm{EII}_{01}$, and implementation details, should accompany the online version of the manuscript.


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## Conflict of interest

The author declares no conflict of interest.

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